

Mitochondrial respiratory supercomplexes in mammalian cells: structural versus functional role.

PMID: 33201259 | Indexed in PubMed

Mitochondria are recognized as the main source of ATP to meet the energy demands of the cell. ATP production occurs by oxidative phosphorylation when electrons are transported through the electron transport chain (ETC) complexes and develop the proton motive force across the inner mitochondrial membrane that is used for ATP synthesis. Studies since the 1960s have been concentrated on the two models of structural organization of ETC complexes known as "solid-state" and "fluid-state" models. However, advanced new techniques such as blue-native gel electrophoresis, mass spectroscopy, and cryogenic electron microscopy for analysis of macromolecular protein complexes provided new data in favor of the solid-state model. According to this model, individual ETC complexes are assembled into macromolecular structures known as respiratory supercomplexes (SCs). A large number of studies over the last 20 years proposed the potential role of SCs to facilitate substrate channeling, maintain the integrity of individual ETC complexes, reduce electron leakage and production of reactive oxygen species, and prevent excessive and random aggregation of proteins in the inner mitochondrial membrane. However, many other studies have challenged the proposed functional role of SCs. Recently, a third model known as the "plasticity" model was proposed that partly reconciles both "solid-state" and "fluid-state" models. According to the "plasticity" model, respiratory SCs can co-exist with the individual ETC complexes. To date, the physiological role of SCs remains unknown, although several studies using tissue samples of patients or animal/cell models of human diseases revealed an associative link between functional changes and the disintegration of SC assembly. This review summarizes and discusses previous studies on the mechanisms and regulation of SC assembly under physiological and pathological conditions.

[Structure, biogenesis and mechanism of function of the mitochondrial ATP synthase complex].

PMID: 23373419 | Indexed in PubMed

Mitochondria are organelles present in all eukaryotic organisms. Their primary function is production of energy in the form of ATP by oxidative phosphorylation. The final step of this process is catalyzed by an enzyme of internal mitochondrial membrane - ATP synthase. The ATP synthase consists of the seventeen subunits in yeast (in vertebrate sixteen is identified to date) organized in hydrophobic, membrane localized unit, referred to as F₀ and hydrophilic domain F₁ directed into mitochondria matrix. Genes encoding the ATP synthase subunits are mainly nuclear, but few of them, encoding hydrophobic subunits, are retained in mitochondrial genome in most Eukaryotes. Biogenesis of the ATP synthase is a sophisticated process, depending on the activity of proteins, which are not ATP synthase subunits, coordinating expression of the nuclear and mitochondrial genes and their assembly in active complex. This review summarizes the present knowledge about structure, biogenesis and mechanism of ATP synthase complex function.

Deep learning-enhanced automated mitochondrial segmentation in FIB-SEM images using an entropy-weighted ensemble approach.

PMID: 39591424 | Indexed in PubMed

Mitochondria are intracellular organelles that act as powerhouses by breaking down nutrition molecules to produce adenosine triphosphate (ATP) as cellular fuel. They have their own genetic material called mitochondrial DNA. Alterations in mitochondrial DNA can result in primary mitochondrial diseases, including neurodegenerative disorders. Early detection of these abnormalities is crucial in slowing disease progression. With recent advances in data acquisition techniques such as focused ion beam scanning electron microscopy, it has become feasible to capture large intracellular organelle volumes at data rates reaching 4Tb/minute, each containing numerous cells. However, manually segmenting large data volumes (gigapixels) can be time-consuming for pathologists. Therefore, there is an urgent need for automated tools that can efficiently segment mitochondria with minimal user intervention. Our article proposes an ensemble of two automatic segmentation pipelines to predict regions of interest specific to mitochondria. This architecture combines the predicted outputs from both pipelines using an ensemble learning-based entropy-weighted fusion technique. The methodology minimizes the impact of individual predictions and enhances the overall segmentation results. The performance of the segmentation task is evaluated using various metrics, ensuring the reliability of our results. We used four publicly available datasets to evaluate our proposed method's effectiveness. Our proposed fusion method has achieved a high score in terms of the mean Jaccard index and dice coefficient for all four datasets. For instance, in the UroCell dataset, our proposed fusion method achieved scores of 0.9644 for the mean Jaccard index and 0.9749 for the Dice coefficient. The mean error rate and pixel accuracy were 0.0062 and 0.9938, respectively. Later, we compared it with state-of-the-art methods like 2D and 3D CNN algorithms. Our ensemble approach shows promising segmentation efficiency with minimal intervention and can potentially aid in the early detection and mitigation of mitochondrial diseases.

Mitochondrial Dysfunctions: Genetic and Cellular Implications Revealed by Various Model Organisms.

PMID: 39336744 | Indexed in PubMed

Mitochondria play a crucial role in maintaining the energy status and redox homeostasis of eukaryotic cells. They are responsible for the metabolic efficiency of cells, providing both ATP and intermediate metabolic products. They also regulate cell survival and death under stress conditions by controlling the cell response or activating the apoptosis process. This functional diversity of mitochondria indicates their great importance for cellular metabolism. Hence, dysfunctions of these structures are increasingly recognized as an element of the etiology of many human diseases and, therefore, an extremely promising therapeutic target. Mitochondrial dysfunctions can be caused by mutations in both nuclear and mitochondrial DNA, as well as by stress factors or replication errors. Progress in knowledge about the biology of mitochondria, as well as the consequences for the efficiency of the entire organism resulting from the dysfunction of these structures, is achieved through the use of model organisms. They are an invaluable tool for analyzing complex cellular processes, leading to a better understanding of diseases caused by mitochondrial dysfunction. In this work, we review the most commonly used model organisms, discussing both their advantages and limitations in modeling fundamental mitochondrial processes or mitochondrial diseases.

Anti-calcitonin gene-related peptide monoclonal antibodies for the treatment of

vestibular migraine: A prospective observational cohort study.

PMID: 36946234 | Indexed in PubMed

Vestibular migraine is considered the most common cause of recurrent vertigo for which specific treatments are missing. Monoclonal antibodies against calcitonin gene-related peptide,, are effective in preventing migraine. Since CGRP is also detected in human cochlear and vestibular organs it may also play a role in vestibular physiology. This is a prospective observational cohort study, aiming at evaluating the efficacy of erenumab, fremanezumab or galcanezumab for the treatment of fifty vestibular migraine patients. We assessed mean monthly days with headache and dizziness/vestibular symptoms, pain intensity and migraine-related clinical burden occurring for 18 months. Response to treatment was excellent as 45 (90%) patients had at least a 50% reduction in vertigo frequency, 43 (86%) had at least a 50% reduction in headache frequency, and 40 (80%) a MIDAS reduction of at least 50%. Overall, 39 (78%) patients had a concomitant reduction of all three parameters. Mean monthly days with dizziness/vestibular symptoms showed an overall significant decrease from a mean of 10.3 ± 1.9 at baseline to 0.8 ± 0.3 days, difference 9.5 (CI 95% 3.6, 15.4; $p < 0.001$) after twelve months. We show that anti-CGRP mAbs may be effective in the treatment of Vestibular Migraine. Their use should be encouraged early in the disease course to allow for a better symptom control and quality of life improvement.

A placebo controlled, randomized clinical trial of galcanezumab for vestibular migraine: The INVESTMENT study.

PMID: 39344988 | Indexed in PubMed

To study if galcanezumab is effective for vestibular migraine (VM). There are currently no placebo-controlled trials showing that treatment is effective for VM. Therefore, we performed the first placebo controlled, randomized clinical trial of a calcitonin gene-related peptide-targeted monoclonal antibody for VM. This was a single site, prospective, double-blind placebo controlled randomized clinical trial. Key inclusion criteria were as follows: participants aged 18-75 years with a diagnosis of VM or probable VM per Barany Society criteria. The primary outcome was change in VM-PATHI (Vestibular Migraine Patient Assessment Tool and Handicap Inventory) score, and secondary outcomes included change in DHI (Dizziness Handicap Inventory) score, and count of definite dizzy days (DDD). Participants were randomized 1:1 to 3 months of treatment with galcanezumab or placebo via subcutaneous injection with a pre-filled syringe, 240 mg the first month, and 120 mg for the second and third months. Forty participants were randomized, and 38 participants were in the modified intent to treat analysis. VM-PATHI score was reduced 5.1 points (95% confidence interval [CI] -13.0 to 2.7) for placebo (N = 21), and 14.8 points (95% CI -23.0 to -6.5) for galcanezumab (N = 17), a difference of -9.6 (95% CI -20.7 to 1.5, $p = 0.044$). DHI dropped 8.3 points in the placebo arm (95% CI -15.0 to 1.6), and 22.0 points in the galcanezumab arm (95% CI -31.9 to -12.1), a difference of -13.7 (95% CI -20.4 to -8.5, $p = 0.018$). The count of DDDs per month dropped from 18 days (standard deviation [SD] 7.6) in the baseline month to 12.5 days (SD 11.2) in month 4 for those in the placebo arm, and from 17.9 days (SD 7.9) in the baseline month to 6.6 days (SD 7.3) in month 4 for those in the galcanezumab arm, a difference of -5.7 days (95% CI -10.7 to -0.7, $p = 0.026$). No serious adverse events were observed. In this pilot study, galcanezumab was effective in treating VM.

Nucleosomal DNA has topological memory.

PMID: 38806488 | Indexed in PubMed

One elusive aspect of the chromosome architecture is how it constrains the DNA topology. Nucleosomes stabilise negative DNA supercoils by restraining a DNA linking number difference (ΔLk) of about -1.26. However, whether this capacity is uniform across the genome is unknown. Here, we calculate the ΔLk restrained by over 4000 nucleosomes in yeast cells. To achieve this, we insert each nucleosome in a circular minichromosome and perform Topo-seq, a high-throughput procedure to inspect the topology of circular DNA libraries in one gel electrophoresis. We show that nucleosomes inherently restrain distinct ΔLk values depending on their genomic origin. Nucleosome DNA topologies differ at gene bodies ($\Delta Lk = -1.29$), intergenic regions ($\Delta Lk = -1.23$), rDNA genes ($\Delta Lk = -1.24$) and telomeric regions ($\Delta Lk = -1.07$). Nucleosomes near the transcription start and termination sites also exhibit singular DNA topologies. Our findings demonstrate that nucleosome DNA topology is imprinted by its native chromatin context and persists when the nucleosome is relocated.

Prenatal Diagnosis of Mucopolysaccharidosis-Plus Syndrome (MPSPS).

PMID: 37628632 | Indexed in PubMed

Mucopolysaccharidosis-plus syndrome (MPSPS) is an autosomal-recessive disorder caused by c.1492C>T (p.R498W) in the VPS33A gene. MPSPS is a severe disorder that causes a short lifespan in patients. Currently, there is no specific treatment for patients. The Yakut population is more prone to this disease than others. Diagnosing MPSPS relies on clinical manifestations, and genetic testing (GT) is used to confirm the diagnosis. In this research, we examined two pregnancy cases, one of which involved a prenatal diagnosis for MPSPS. Notably, neither pregnant woman had a known family history of the disorder. During their pregnancies, both women underwent prenatal ultrasonography, which revealed increased prenatal thickness during the second trimester. In the first case, ultrasonography indicated increased prenatal thickness in the second trimester, but a definitive diagnosis was not made at that time. The patient was eventually diagnosed with MPSPS at 11 months of age. On the contrary, in the second case, GT uncovered that the parents were carriers of MPSPS. Consequently, a placental biopsy was performed, leading to an early diagnosis of MPSPS. This study emphasizes the importance of ultrasonography findings in prenatal MPSPS diagnosis. Combining ultrasonography with GT can be a valuable approach to confirming MPSPS at an early stage, allowing for the appropriate planning of delivery methods and medical care. Ultimately, this comprehensive approach can significantly enhance the quality of life of both affected patients and their parents.

Mucopolysaccharidosis-Plus Syndrome, a Rapidly Progressive Disease: Favorable Impact of a Very Prolonged Steroid Treatment on the Clinical Course in a Child.

PMID: 35327996 | Indexed in PubMed

Mucopolysaccharidosis-plus syndrome (MPS-PS) is a novel autosomal recessive disorder caused by a mutation in the VPS33A gene. This syndrome presents with typical symptoms of mucopolysaccharidosis, as well as congenital heart defects, renal, and hematopoietic system disorders. To date, twenty-four patients have been described. There is no specific therapy for MPS-PS; clinical management is therefore

limited to symptoms management. The clinical course is rapidly progressive, and most patients die before 1-2 years of age. We describe a currently 6-year-old male patient with MPS-PS presenting with multiorgan involvement. Symptoms started at four months of age when he progressively suffered from numerous acute and potentially life-threatening events. When he was two years old, he developed secondary hemophagocytic lymphohistiocytosis (HLH), which was successfully treated with steroids. To date, this child represents the oldest patient affected by MPS-PS described in the literature and the first one presenting with a life-threatening secondary HLH. The prolonged steroid treatment allowed a stabilization of his general and hematological conditions and probably determined an improvement of his psychomotor milestones and new neurological acquisitions with an improvement of quality of life. HLH should be suspected and adequately treated in MPS-PS patients presenting with suggestive symptoms of the disease. The usefulness of a prolonged steroid treatment to improve the clinical course of children with MPS-PS deserves further investigation.

Mucopolysaccharidosis-Plus Syndrome: Is This a Type of Mucopolysaccharidosis or a Separate Kind of Metabolic Disease?

PMID: 39273517 | Indexed in PubMed

Several years ago, dozens of cases were described in patients with symptoms very similar to mucopolysaccharidosis (MPS). This new disease entity was described as mucopolysaccharidosis-plus syndrome (MPSPS). The name of the disease indicates that in addition to the typical symptoms of conventional MPS, patients develop other features such as congenital heart defects and kidney and hematopoietic system disorders. The symptoms are highly advanced, and patients usually do not survive past the second year of life. MPSPS is inherited in an autosomal recessive manner and is caused by a homozygous-specific mutation in the gene encoding the VPS33A protein. To date, it has been described in 41 patients. Patients with MPSPS exhibited excessive excretion of glycosaminoglycans (GAGs) in the urine and exceptionally high levels of heparan sulfate in the plasma, but the accumulation of substrates is not caused by a decrease in the activity of any lysosomal enzymes. Here, we discuss the pathomechanisms and symptoms of MPSPS, comparing them to those of MPS. Moreover, we asked the question whether MPSPS should be classified as a type of MPS or a separate disease, as contrary to 'classical' MPS types, despite GAG accumulation, no defects in lysosomal enzymes responsible for degradation of these compounds could be detected in MPSPS. The molecular mechanism of the appearance of GAG accumulation in MPSPS is suggested on the basis of results available in the literature.